

5.4.6

EE25BTECH11011 - Navya

Question:

Using elementary transformations, find the inverse of the following matrix.

$$\begin{pmatrix} 10 & -2 \\ -5 & 1 \end{pmatrix}$$

Solution:

We employ the Gauss–Jordan method,

$$\left(\begin{array}{cc|cc} 10 & -2 & 1 & 0 \\ -5 & 1 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow \frac{1}{10}R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{5} & \frac{1}{10} & 0 \\ -5 & 1 & 0 & 1 \end{array} \right) \quad (1)$$

$$\xrightarrow{R_2 \leftarrow R_2 + 5R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{5} & \frac{1}{10} & 0 \\ 0 & 0 & \frac{1}{2} & 1 \end{array} \right) \quad (2)$$

At this stage, the left-hand side is not the identity matrix.
Hence the given matrix is not invertible.